Flood situation report

Area: Melamchi flood, Nepal, 15 Jun 2021 | Event date: 15 June 2021 | Latest radar image: 27 June 2021

1. Where it flooded

Satellite images show 139 ha (1.39 km²) flooded or buried by debris. F010 F011

Of this, 22 ha is open water and 117 ha is debris or scour. F012 F014

The radar could not see 1,482 ha because of terrain shadow, so conditions there are unknown. F016

The places with the most buildings likely destroyed: Melamchi, Bansbari, Sikharpur, Kharchung. F049



Radar change map, before and after the event.

2. What was hit

Buildings: 150 likely destroyed and 69 possibly damaged, out of 41,988 mapped. F025 F021 F022

Roads: 6.7 km of road lies under mapped debris or water; the road sections that touch it total 18.6 km, out of 1,881.9 km mapped. F040 F034 F038

Bridges: 5 likely destroyed (Bhothang Road) and 1 possibly damaged, out of 80 mapped. F031 F027 F028 F033

3. Who is cut off

1 of 58 settlements are likely cut off by road. F051 F050

Among them: Melamchi. F055

The nearest health facility for 1 of the cut-off settlements is Lagarche Health Post. F056 F057

OpenStreetMap lists 4 health facilities in the area (hospitals, health posts and clinics); its labels are not verified. F430 F431

Repair priority

1. Reopening Melamchi Road reconnects 1 settlement. F433 F434 F436

Method

Sentinel-1 radar images from before and after the event are compared to map new water and debris. Pre-event OpenStreetMap roads, bridges and buildings are overlaid, and a road network model finds settlements that can no longer reach a health facility.

Educational prototype, not an operational tool. Numbers come from satellite analysis and have not been checked on the ground.

Grey codes such as F012 point to entries in facts.json.

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